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MID-SEMESTER EXAMINATION – SEPTEMBER 2025

Name of the Course: B.Tech-ECE

Name of the Paper: Electronics Engineering Materials

Time: 1.5 Hours

Semester: III

Paper Code: TEC 343

Maximum Marks: 50

Note:-

- (i) All questions are compulsory.
- (ii) Answer any one sub questions among all five questions.
- (iii) Each question carries 10 marks.

Q1	(10marks)	CO1
(a)	State interference phenomenon and explain Bragg's diffraction.	
OR		
(b)	State and explain various types of point defects in crystal.	
Q2	(10 marks)	CO2
(a)	Find the conductivity of silicon (i) under intrinsic condition at room temperature (ii) with donor impurity of 1 part in 5×10^7 . Given that, intrinsic concentration = $1.5 \times 10^{10}/cm^3$, mobility of electrons = $1300 cm^2/Vs$, mobility of holes = $500 cm^2/Vs$, number of silicon atoms = $5 \times 10^{22}/cm^3$ and charge of an electron = $1.6 \times 10^{-19} C$.	
OR		
(b)	Describe a technique for the growth of single crystal and the zone refining.	
Q3	(10 marks)	CO1 and CO2
(a)	Explain edge and screw dislocation with diagrams mentioning the associated Berger Vector.	
OR		
(b)	Calculate (i) Relaxation time (ii) Resistivity of conductor (iii) Velocity of electron with the Fermi energy for a known semiconductor with Fermi energy = $5.5 eV$, mobility of electrons = $7.04 \times 10^{-3} m^2/Vs$, number of conduction electrons = $5.8 \times 10^{28} m^3$, charge of an electron = $1.6 \times 10^{-19} C$, mass of an electron = $9.1 \times 10^{-31} kg$.	
Q4	(10 marks)	CO2 and CO3
(a)	Consider the MOS structure that consists of a P type doped Si substrate, a SiO_2 layer and a metal gate (Al). The equilibrium Fermi potential of the doped Si substrate is $0.2 eV$. Using electron affinity for Si ($4.15 eV$), work function for Al ($4.1 eV$) and bandgap for Si ($1.1 eV$), calculate the work function difference across the MOS system?	
OR		
(b)	Derive a relation between intensity of polarization (P) and electric field (E). When NaCl crystal is subjected to an electric field of $50 V/cm$, the resulting polarization is $2.215 \times 10^{-7} C/m^2$. Calculate relative permittivity of NaCl.	
Q5	(10 marks)	

(a)	Derive the percentage packing fraction of a Body centered cubic crystal (BCC) structure.	CO2, CO3
OR		
(b)	Explain electronic polarization. Calculate the electronic polarizability of argon atom. Given $\epsilon_r = 1.0024$ and number of atoms per unit volume, $N = 2.7 \times 10^{25} \text{ atoms/m}^3$.	